

# MATH 330 Homework 7

## Cardinality

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**Exercise 1.** Show that if  $A \subseteq B$ , then  $A \preceq B$ .

**Exercise 2.** Prove that the relation  $\prec$  is transitive.

**Exercise 3.** Prove that a subset of a finite set is finite.

**Exercise 4.** Let  $A$  and  $B$  be finite sets. Prove:

1.  $A \cap B$  is finite.
2.  $A \cup B$  is finite.

**Exercise 5.** Let  $A$  be finite and let  $B \subsetneq A$ .

1. Prove that  $B \preceq A$ .
2. Prove that  $A$  and  $B$  are not equinumerous.

**Exercise 6.** Prove that a countable set with an infinite subset is denumerable.

**Exercise 7.** Let  $\mathbb{F} = \mathbb{N} \times \mathbb{N}$ , and write a pair  $(a, b)$  as the formal fraction  $a/b$ . Show that

$$f : \mathbb{F} \rightarrow \mathbb{N} \times \mathbb{N}, \quad f(a/b) = (a, b),$$

is a bijection.

**Exercise 8.** Assume the Axiom of Choice. Let  $A$  and  $B$  be nonempty sets. Prove that there is an injection  $A \rightarrow B$  if and only if there is a surjection  $B \rightarrow A$ .

**Exercise 9.** Prove that each set is denumerable.

1. The nonnegative integers  $\mathbb{N} \cup \{0\}$ .
2. The integers  $\mathbb{Z}$ .

**Exercise 10.** Prove:

1. The union of two countable sets is countable.
2. The union of finitely many countable sets is countable.

**Exercise 11.** Let  $A$  and  $B$  be denumerable. Prove that  $A \times B$  is denumerable.

**Exercise 12.** Let  $A, B, C$  be sets and define  $A \times B \times C = \{(a, b, c) \mid a \in A, b \in B, c \in C\}$ . Prove that  $A \times B \times C \equiv (A \times B) \times C$ .

**Exercise 13.** For  $n \in \mathbb{N}$ , let  $\mathbb{N}^n$  be the set of ordered  $n$ -tuples of natural numbers. Prove that  $\mathbb{N}^n$  is countable.

**Exercise 14.** Let  $\mathbb{S}$  be the set of all infinite sequences of zeros and ones. Prove that  $\mathbb{S}$  is uncountable.